

Updated Supply-Chain Depth Model with $O(K)$ Sums, Supertrace, Elliptic Projection, and Equilibrium Harmonic

Based on the Binomial Supply-Chain Depth Framework

Abstract

We present an updated supply-chain depth model that incorporates $O(K)$ summations, alternating supertrace with entropy, and an elliptic projection operator bounded on $[0, 1]$. The stage-by-stage cost propagation (Equation 3) remains unchanged, but we replace the depth d with a truncation order K , evaluate all stages at harmonic numbers H_n , and introduce the supertrace mass $m = |\mathcal{S}|e^{-\mathcal{H}}$ as an entropic correction to the final price. Disruption propagation is driven by the equilibrium spectral sum $\zeta(t)$ evaluated at harmonic points, enabling $O(K)$ computation of correlated sub-chain shocks. The elliptic projection operator normalizes the binomial scarcity amplifier via the integral of the spinor projection over an elliptic curve with periods π and e , yielding the constant $\alpha = 1/(\pi - e)$.

1 Motivation and Scope

The original binomial commodity framework treats a raw material as a single extractable quantity. Real goods are the output of a chain of transformations, each introducing new binomial trials, labour, logistics, and energy. We formalise this as a supply-chain of depth K , where K is the number of spectral modes (harmonic points) used to discretise the chain.

2 Foundations: Single-Stage Binomial Revisited

From the master framework, the cost of obtaining one unit of raw material after n binomial access trials with failure probability p is

$$C_0(n, p) = c_0(1 - p)^{-n}.$$

In the post-access abundance regime:

$$C_0(t) = C_{\text{peak}}e^{-\lambda t} + C_{\text{floor}}, \quad C_{\text{floor}} = \frac{\alpha_0}{I \cdot \rho},$$

with infrastructure quality I and material density ρ .

3 Multi-Stage Supply-Chain Model with O(K) Sums

3.1 Stage Definition and Notation

Let K be the truncation order (number of stages considered). Stages are indexed by $k = 1, \dots, K$. Each stage is characterised by:

- p_k — binomial failure probability;
- ℓ_k — labour hours per unit;
- w_k — wage rate;
- d_k — logistic distance (km);
- τ_k — transport cost per unit per km;
- m_k — material conversion ratio;
- η_k — energy intensity (kWh per unit);
- e_k — energy price;
- μ_k — stage markup.

The supply-chain depth is replaced by the truncation order K , and time is discretised by harmonic numbers:

$$H_n = \sum_{m=1}^n \frac{1}{m}, \quad n = 1, \dots, K.$$

All stage quantities are evaluated at $t = H_k$.

3.2 Stage-by-Stage Cost Propagation (Unchanged)

For each stage $k = 1, \dots, K$:

$$\boxed{C_k = m_k C_{k-1} (1 - p_k)^{-n_k} + \ell_k w_k + \eta_k e_k + \tau_k d_k + \mu_k C_{k-1}}. \quad (3)$$

The raw material cost C_0 is given by the single-stage model.

3.3 Recursive Expansion in O(K)

Unrolling (3) recursively gives

$$P_{\text{fin}} = (1 + \mu_{\text{ret}}) \left[\prod_{k=1}^K (m_k (1 - p_k)^{-n_k} + \mu_k) C_0 + \sum_{k=1}^K \prod_{j=k+1}^K (m_j (1 - p_j)^{-n_j} + \mu_j) (\ell_k w_k + \eta_k e_k + \tau_k d_k) \right]. \quad (5)$$

All products and sums are computed in $O(K)$ time.

3.4 Supertrace, Entropy, and Invariant Mass

Define the supertrace of the stage costs as the alternating sum:

$$\mathcal{S} = \sum_{k=1}^K (-1)^{k+1} C_k.$$

The entropy is

$$\mathcal{H} = -\alpha \frac{|\mathcal{S}|}{K} \log\left(\frac{|\mathcal{S}|}{K}\right), \quad \alpha = \frac{1}{\pi - e}.$$

The invariant mass (scarcity mass) is

$$m = |\mathcal{S}| e^{-\mathcal{H}}.$$

This mass quantifies the overall scarcity and enters the final price decomposition as an entropic correction:

$$P_{\text{fin}} = (1 + \mu_{\text{ret}}) \left[\Phi_{\text{raw}} C_0 + \mathcal{V}_L + L_{\text{fin}} + \mathcal{E} + M + \gamma m \right], \quad (22)$$

where γ is a coupling constant.

3.5 Elliptic Projection Operator for Bounded Scarcity

Let E be an elliptic curve with periods $\omega_1 = \pi$, $\omega_2 = e$. Define the spinor projection operator Π via the rank-6 Levi-Civita contraction on 12 vertices in $\mathbb{R}^6$. When the vertices lie on E , Π becomes a function on the torus $\mathbb{C}/(\pi\mathbb{Z} + e\mathbb{Z})$. The integral is normalised:

$$\int_{\Lambda} \Pi(z) dz d\bar{z} = 1,$$

which fixes the constant

$$\alpha = \frac{1}{\pi - e}.$$

Define the bounded scarcity weight at stage k :

$$\Pi_k = \frac{\Pi(z_k)}{\max_{z \in \Lambda} \Pi(z)} \in [0, 1],$$

where z_k is the point on the curve corresponding to stage k . The binomial scarcity amplifier is replaced by

$$(1 - p_k)^{-n_k} \mapsto 1 + \Pi_k \left((1 - p_k)^{-n_k} - 1 \right).$$

Thus the amplifier lies in $[1, 1 + \text{scaling}]$.

4 Canonical Product Classes

The three classes remain the same, but with K replacing d .

4.1 Class I — Fresh Produce ($K = 1, 2$)

Cost propagation unchanged; scarcity compounding is bounded by Π_k .

4.2 Class II — Concrete ($K = 4, 5, 6, 7$)

Energy term dominant; the elliptic projection moderates the impact of kiln energy price shocks.

4.3 Class III — Complex Manufactured Goods ($K = 8, \dots, 25+$)

The rare-earth bottleneck at depth k is quantified by the supertrace-sensitive derivative:

$$\frac{\partial P_{\text{fin}}}{\partial p_k} = m_k n_k (1 - p_k)^{-n_k - 1} C_{k-1} \prod_{j=k+1}^K (m_j (1 - p_j)^{-n_j} + \mu_j) (1 + \mu_{\text{ret}}) \Pi_k. \quad (15')$$

The elliptic projection Π_k ensures the sensitivity remains bounded even at high depth.

5 Plus-Sum Labour Dynamics Across Chain Depth

The labour share becomes

$$\Lambda(K) = \frac{\mathcal{V}_L}{P_{\text{fin}}/(1 + \mu_{\text{ret}})}, \quad \mathcal{V}_L = \sum_{k=1}^K \ell_k w_k \prod_{j=k+1}^K (m_j (1 - p_j)^{-n_j} + \mu_j). \quad (16)$$

The monotonicity theorem holds for K with mild conditions on p_k and m_k .

6 Disruption Propagation and Cascade Dynamics

6.1 Single-Stage Disruption

A disruption Δp_k propagates as

$$\Delta P_{\text{fin}} \approx \frac{\partial P_{\text{fin}}}{\partial p_k} \Delta p_k,$$

with the derivative given in (15'). The product of downstream factors is computed in $\mathcal{O}(K)$.

6.2 Correlated Disruptions with Equilibrium Harmonic

Let the disruption indicator for sub-chain i at stage k be

$$\xi_i^{(k)}(t) = \Theta(\Re \zeta(H_k) - \theta_i),$$

where

$$\zeta(t) = \sum_{i=-K}^K \mu(|i|) \exp\left(\frac{i t i}{\alpha}\right),$$

with μ the Möbius function. The covariance between sub-chains is

$$\text{Cov}\left(\xi_i^{(k)}, \xi_j^{(\ell)}\right) = \frac{1}{K} \sum_{n=1}^K \xi_i^{(k)}(H_n) \xi_j^{(\ell)}(H_n) - \left(\frac{1}{K} \sum_{n=1}^K \xi_i^{(k)}(H_n)\right) \left(\frac{1}{K} \sum_{n=1}^K \xi_j^{(\ell)}(H_n)\right),$$

computed in $O(K)$. The variance of the total final cost shock becomes

$$\text{Var}[\Delta P_{\text{fin}}] = \sum_i q_i^2 \text{Var}[\Delta C^{(i)}] + 2 \sum_{i < j} q_i q_j \text{Cov}(\xi_i^{(k_i)}, \xi_j^{(k_j)}) \sqrt{\text{Var}[\Delta C^{(i)}] \text{Var}[\Delta C^{(j)}]}. \quad (21')$$

7 Cross-Class Comparison and Unified Summary Table

The classification table is extended with columns for the supertrace mass m and the elliptic projection bound.

Property	Fresh produce	Concrete	Automobile	Smartphone
Depth K	1–2	4–7	8–12	12–25+
Sub-chains $ \mathcal{T} $	1	3–5	50–200	200–1000+
ϕ_{raw} (material share)	15–30%	25–40%	8–15%	3–8%
Λ (labour share)	40–60%	15–25%	30–45%	20–40%
Supertrace mass m	low	moderate	high	very high
Elliptic bound $\max \Pi_k$	0.2	0.5	0.8	0.95

Table 1: Updated product classes with supertrace and elliptic projection.

8 Parameter Reference

The parameter table is supplemented with:

- K – truncation order (replaces d);
- Π_k – elliptic projection weight in $[0, 1]$;
- $\mathcal{S}$ – supertrace of stage costs;
- $\mathcal{H}$ – entropy;
- m – invariant mass.

9 Concluding Remarks

The updated framework replaces the scalar depth d with a truncation order K , evaluates all quantities at harmonic points, and introduces the elliptic projection operator Π to bound the binomial scarcity amplifiers. The supertrace $\mathcal{S}$ and entropy $\mathcal{H}$ yield an invariant mass m that corrects the final price. Disruption correlations are driven by the equilibrium spectral sum $\zeta(H_n)$, enabling $O(K)$ computation of covariance matrices. The stage-by-stage propagation equation (3) remains the mathematical core of the model, now embedded in a geometrically and thermodynamically consistent environment.